Project 1 – Railways Gauges

The task is to load the railway gauges csv file to pandas data frame, read the dataset, find the maximum installations from the dataset and provide the pictorial view of all data using bar chart.

When working with tabular data, such as data stored in spreadsheets or databases, pandas is one of the right tool. Pandas is used to explore, clean, and process the data. In pandas, a data table is called a DataFrame.

In order to achieve the tasks first we need to import the given CSV file to pandas library.

To import railway gauges file to pandas we can use read_csv('File_name.csv') function as shown below.

```
#loading csv file
df=pd.read_csv('railway_gauges.csv')
```

In the above snippet df is the variable name for storing and accessing dataset. pd is the alias name used to access pandas module. read_csv is the panda function for reading the railway_gauges file.

Once the file is loaded to the pandas library, we need to print the first or top rows from the dataset. To achieve this we are using pandas builtin function head(). This function will provide the top 5 rows along with the column names. And also we have tail() function which will provide the last 5 rows from an given dataset.

To print the top 5 rows we can use the code as shown below.

```
#printing top 5 rows along with column names
print("Top 5 Rows are\n",df.head(),'\n')
```

Here we are printing the top 5 rows of the dataset using df.head() function. Where df is the variable name which was used while loading the csv file to the pandas library.

After executing the above code, we will receive the output as shown below.

```
Top 5 Rows are
       Year  Broad Gauge  Metre Gauge  Narrow Gauge  Total
0  1964-65         3521         2891           464   6876
1  1965-66         3603         2913           470   6986
2  1966-67         3615         2911           481   7007
3  1967-68         3601         2920           506   7027
4  1968-69         3603         2921           508   7032
```

The next task is to identify the year which has maximum installations.

In pandas library we do have so many properties and functions to achieve certain tasks. Some are mentioned below.

iloc -> This is one of the pandas property which is used to select/access a DataFrame or dataset or series using their indexes(0-based indexing).

> For example, to access the data from 1st row and 2nd column from an dataset named df, we can write the code as df.iloc[0,1].

loc -> This property is used to select the data from a dataset using their labels or names.

> For example, if we want to access the data of virat's python marks from a marks table, we can directly provide the row name and column name as df.loc['virat','python_marks']. Where row name is virat and column name is marks.

max() -> This is one of the pandas function which is used to get the highest value from a dataset.

> For example, if we run print(df.max()), if the dataset contain single column it will provide single value. And if the dataset df contain more than one column it wil provide single maximum values for each columns in a list. To get values from rows, print(df.max(axis=1))

Idxmax() -> This function is used to get the highest value location(instead of value) from a given dataset.

> For example, if we run print(df.idxmax()) by default it will check for each column with highest value and prints the row name which have highest value. If we run df.idxmax(axis=1) it will check each rows and prints the column name with highest value.

To find the year which have most installations we can use the code shown below.

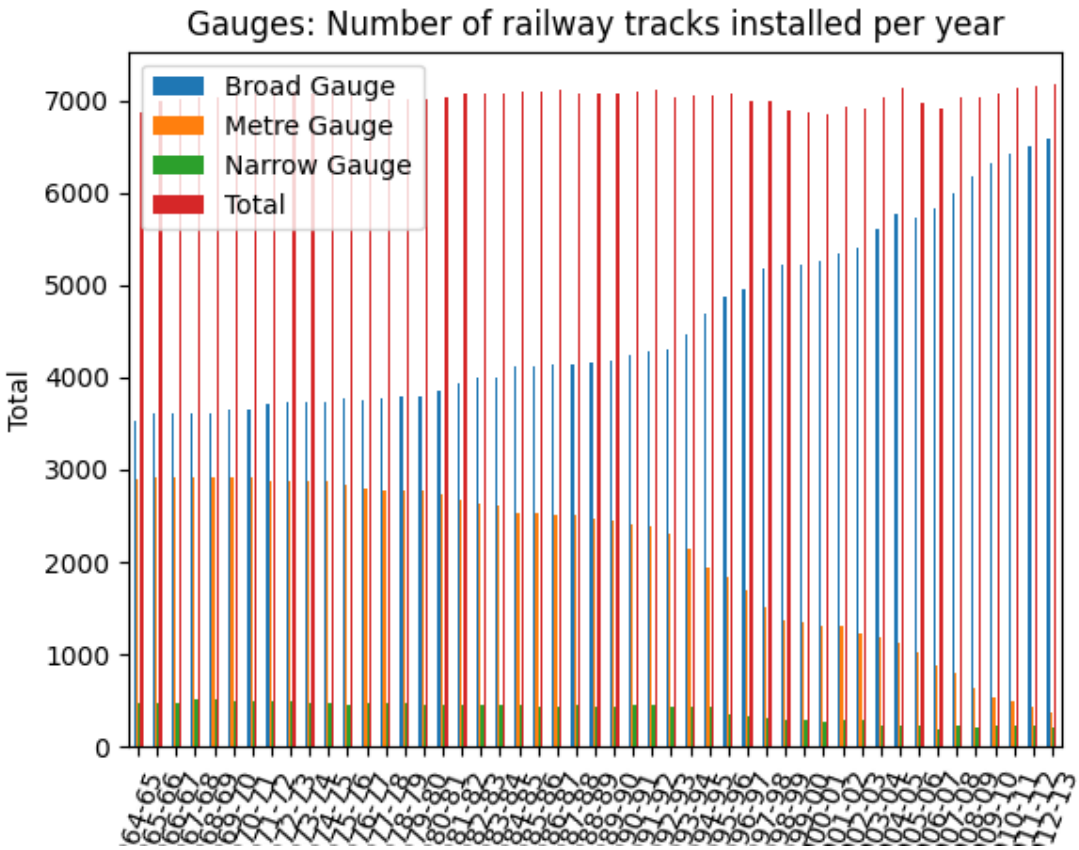
```
#finding which year has maximum installations
print(df.iloc[[df['Total'].idxmax()]])
```

This code will check in the Total column for the hhighest value and provide the entire row details along with the column names.

```
       Year  Broad Gauge  Metre Gauge  Narrow Gauge  Total
48  2012-13         6577          379           216   7172
```

Data Visualization.

We can graphically represent the given dataset using matplotlib module. It provides so many functions like plot(), label(), savefig() etc.. To provide the pictorial view of given rail gauge data set we can use the following code.

```
#plot data using bar chart
ax=df.plot(x='Year',kind='bar')
plt.xticks(rotation=70)
plt.xlabel('Year')
plt.ylabel('Total')
plt.title('Gauges: Number of railway tracks installed per year')
plt.savefig('Graphs/rail_gauges.png')
plt.show()
```

The bar chart of the above code is as follows.